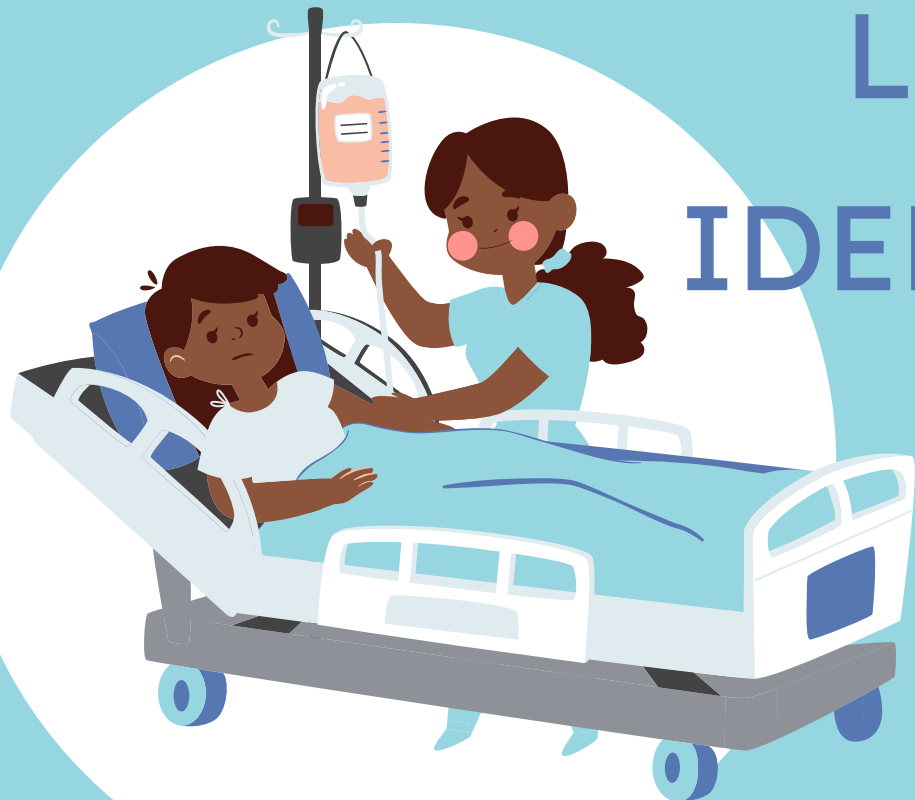


LUNG CANCER IDENTIFICATION



MSAI 495: Special Topics - Deep Learning
[Chukwueloka Obi]

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01

INTRODUCTION

Present our theme and motivation behind the project



Problem Statement and Project Theme

- With Lung Cancer being the main cause of death related to cancer we found that the current decision-making process is insufficient in helping medical practitioners make sound decisions.
- We were motivated to work towards this dataset to glean insights on the Lung Cancer detection methodologies by reading the Machine Learning Models For Classification of Lung Cancer and Selection of Markers Paper

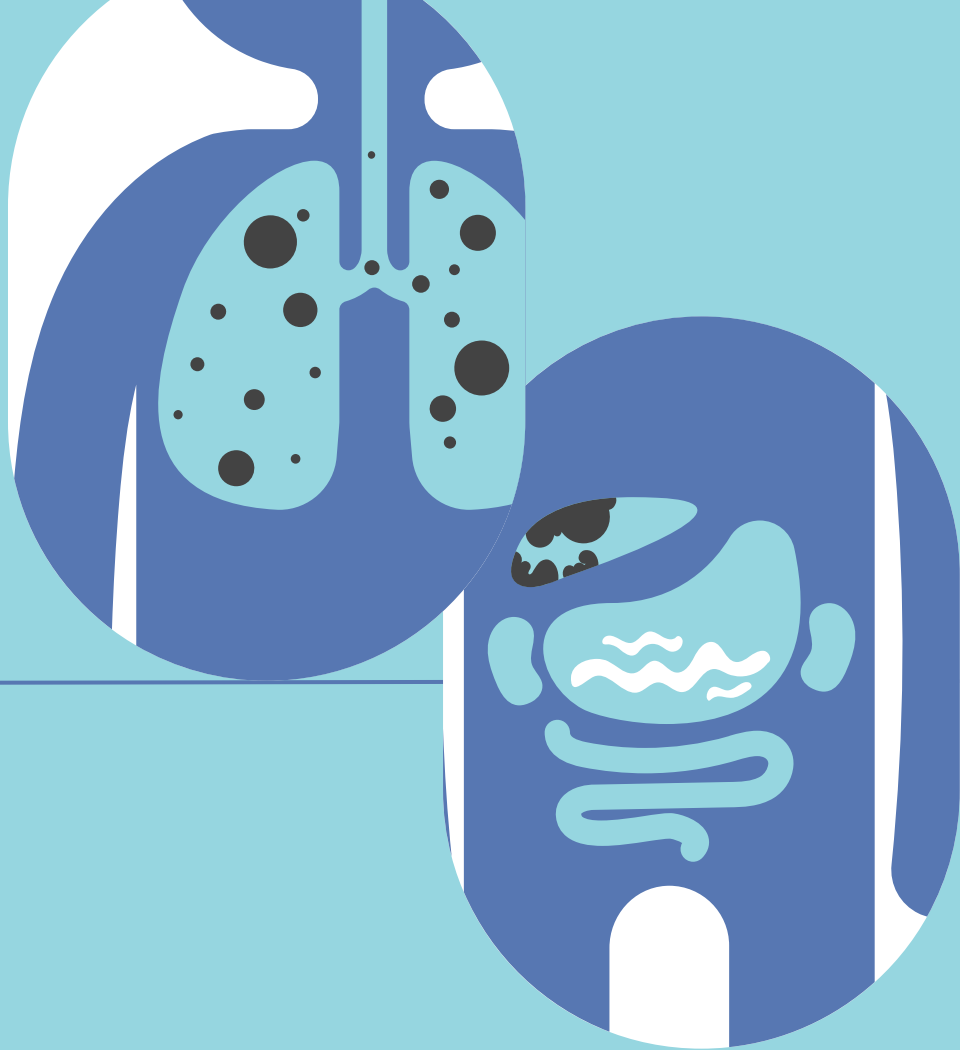
Related Research

- There have been many approaches in trying to accurately recognize lung cancer and its complexities but there are limits to the current procedures that we currently have.
- We now have new clinical data assessment tools that are needed to supplement biopsy and help better identify disease features.
- The previous SVM was thought to be needed to be taken up a notch to a CNN because it was thought to be more effective with the Image Dataset. This belief only grew when we read the Eldridge paper that showed the distinct biologic and diagnostic patterns with tumor microanatomy.

02

DATASET DESCRIPTION

Complete breakdown of dataset and walk-through of analysis



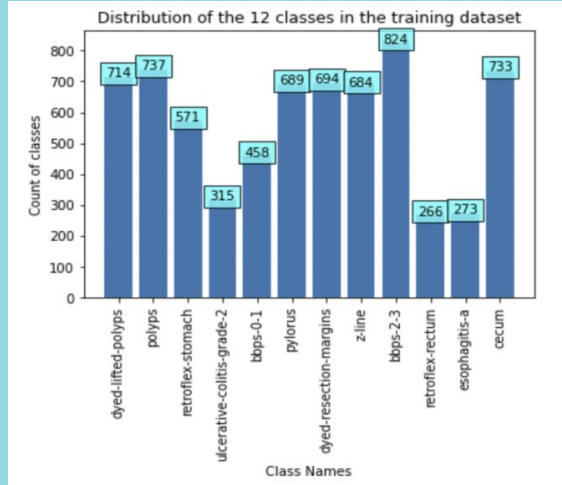
Lung Cancer Image Dataset collected from Kaggle comprising 9717 files

Training set	Validation set	Test set
6958	1956	938

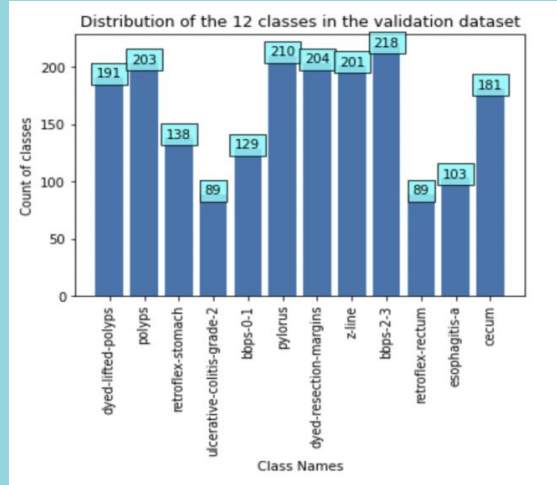
We have a test set, training set, and a validation set. Each of these sets have 12 directories filled with images that could potentially show the stages of lung cancer. We have images of the stomach, esophagus, and bowels.

- Bbps-0-1
- Bbps-2-3
- Cecum
- Ulcerative-colitis-grade-2
- Dyed-resection-margins
- Pylorus
- Z-line
- Retroflex-stomach
- Esophagitis-a
- Retroflex-rectum
- Polyps
- Dyed-lifted-polyps

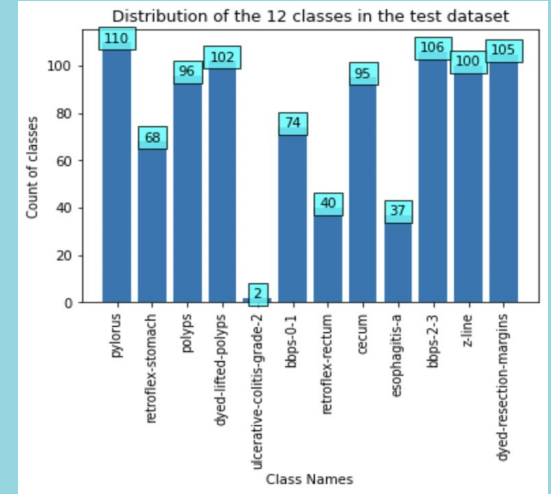
Class Distribution



Class Description in Train Dataset

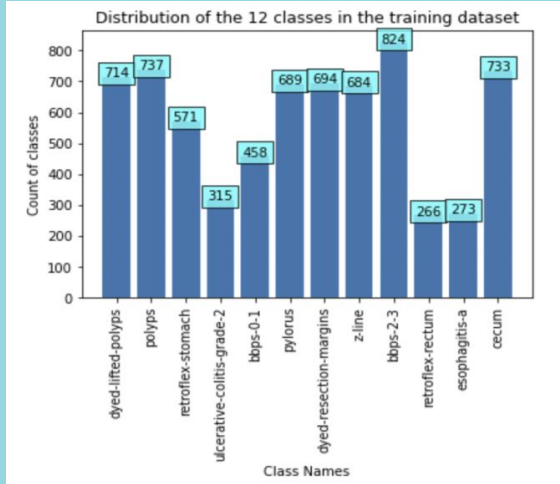


Class Description in Valid Dataset



Class Description in Test Dataset

Class Weights



Class Description in Train Dataset

```
{0: 1.0733333333333333,  
1: 0.8586666666666667,  
2: 0.8288288288288288,  
3: 1.4311111111111111,  
4: 0.8586666666666667,  
5: 0.8434839554682384,  
6: 0.8586666666666667,  
7: 0.8552456839309429,  
8: 2.1466666666666665,  
9: 2.1466666666666665,  
10: 0.8484848484848485,  
11: 0.8304319793681496}
```

Class weights for sample dataset

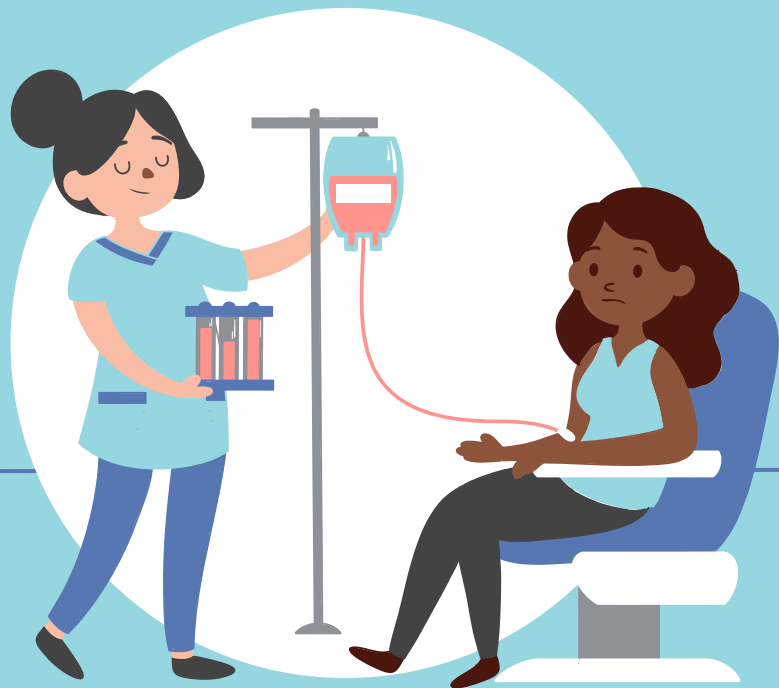
```
{0: 0.8287900526064084,  
1: 1.0352449223416966,  
2: 0.7880854934060937,  
3: 0.8090569561157797,  
4: 1.8338624338624339,  
5: 1.2894345238095237,  
6: 2.204834605597964,  
7: 0.7838082315694256,  
8: 2.1964512040557667,  
9: 0.6804083235178642,  
10: 0.880589430894309,  
11: 0.8252380952380952}
```

Class weights for complete dataset

03

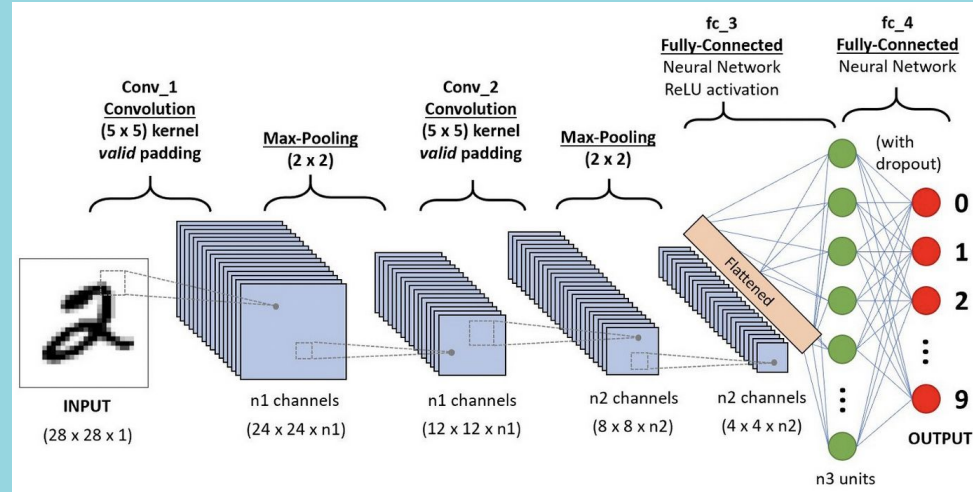
MODEL SPECIFICATION

Model definition and implementation results



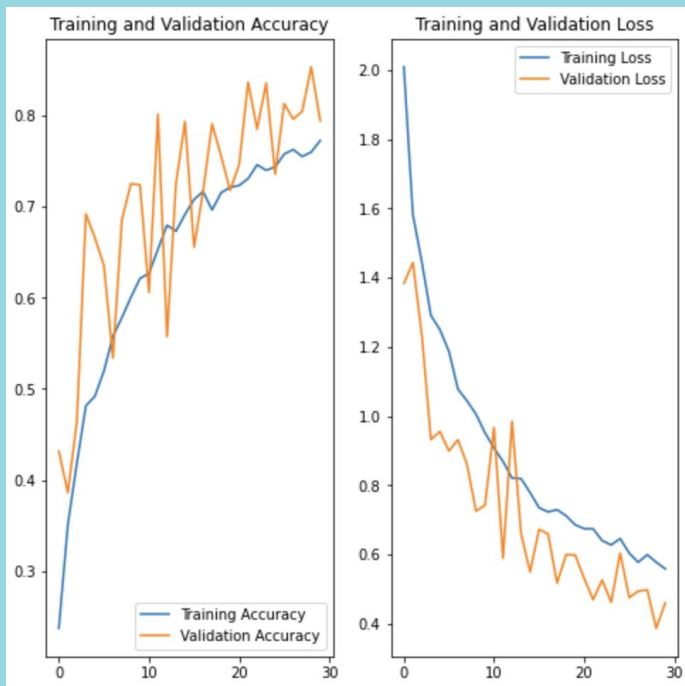
Approach 1: Convolutional Neural Networks (with sample of data)

- A 5-layered CNN architecture is designed for a small sample of images taken from our dataset. (Image included for reference).
- Our CNN model has 5 convolutional layers each followed by a max-pooling layer which is finally flattened into a dense layer to feed into the final output layer with 12 neurons (our 12 classes).
- RMSProp optimizer with a learning rate of 0.001 is used in the architecture.



(Source):
<https://towardsdatascience.com/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way-3bd2b1164a53>

Approach 1: [Contd.]



Training and Validation accuracy and loss plot

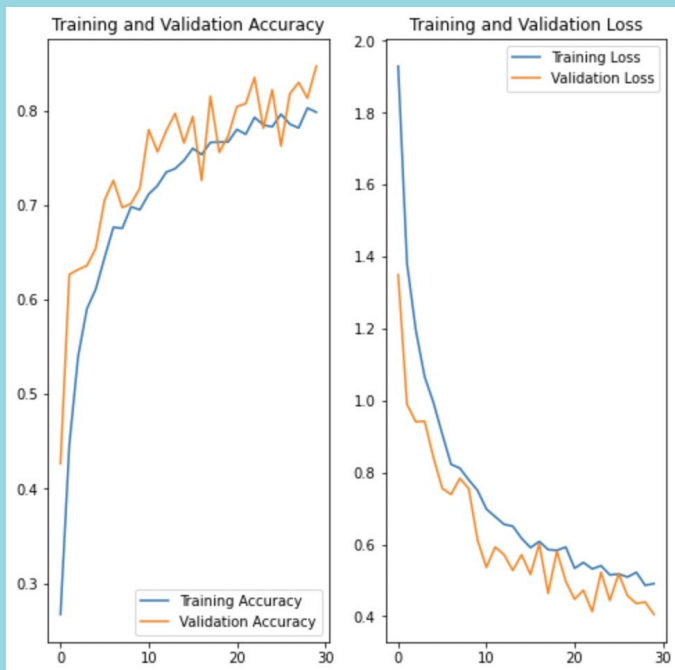
	precision	recall	f1-score	support
0	0.97	0.92	0.94	74
1	0.95	1.00	0.98	106
2	0.89	0.94	0.91	95
3	0.67	0.62	0.64	26
4	0.75	0.42	0.54	105
5	0.97	0.93	0.95	105
6	0.88	0.40	0.55	109
7	0.97	0.97	0.97	68
8	0.32	0.81	0.46	37
9	0.89	0.95	0.92	41
10	1.00	0.89	0.94	96
11	0.60	0.84	0.70	102
micro avg	0.81	0.80	0.81	964
macro avg	0.82	0.81	0.79	964
weighted avg	0.85	0.80	0.80	964
samples avg	0.80	0.80	0.80	964

Precision, Recall, and F-1 scores on test set

Approach 2: Convolutional Neural Networks (with complete data)

- We worked with a similar architecture with minor changes added -
 1. Using the entire dataset instead of a small sample
 2. An additional dense layer added before the output layer to condense more information instead of directing information from 128 neurons directly into the 12 output neurons.
 3. Changed the optimizer function from RMSprop to Adam for the same learning rate.
- The updated model did improve the overall result as compared to our previous attempt.

Approach 2: [Contd.]



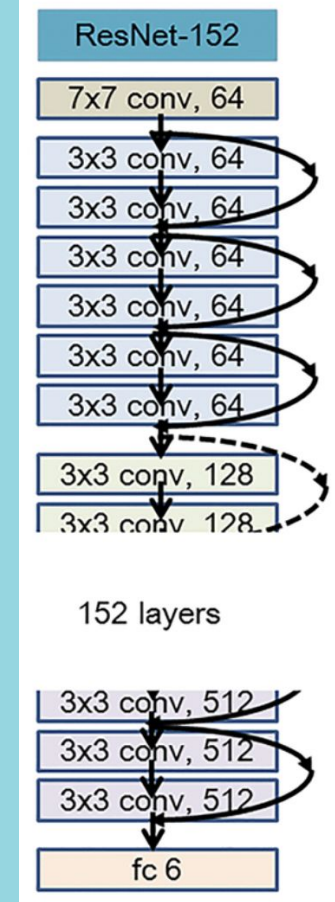
Training and Validation accuracy and loss plot

	precision	recall	f1-score	support
0	0.89	0.95	0.92	110
1	0.98	0.90	0.94	68
2	1.00	0.94	0.97	96
3	0.67	0.59	0.62	102
4	0.14	1.00	0.25	2
5	0.94	0.91	0.92	74
6	0.78	1.00	0.88	40
7	0.98	0.93	0.95	95
8	0.50	0.38	0.43	37
9	0.98	0.97	0.98	106
10	0.79	0.61	0.69	100
11	0.68	0.66	0.67	105
micro avg	0.85	0.81	0.83	935
macro avg	0.78	0.82	0.77	935
weighted avg	0.85	0.81	0.83	935
samples avg	0.81	0.81	0.81	935

Precision, Recall, and F-1 scores on test set

Approach 3: Resnet-152

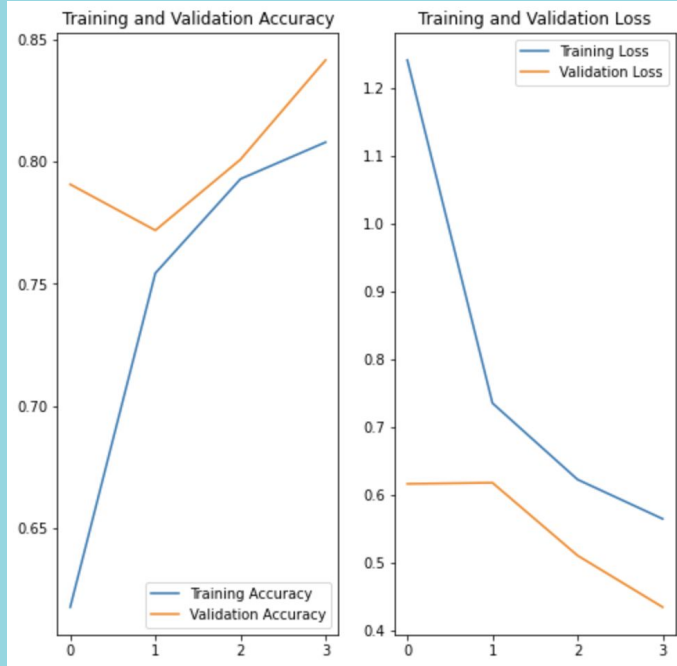
- Resnet-152 is a pre-trained model, and as the name suggests with 152 layers.
- What makes Resnet special? - Presence of residual block in the form of identity connection to tackle the problem of vanishing gradients in backpropagation of deep networks.
- For our implementation, we use pre-trained weights and train only on the last few layers to achieve a classification model catered to our dataset.



(Source):

https://www.researchgate.net/figure/The-representation-of-model-architecture-image-for-ResNet-152-VGG-19-and-two-layered_fig2_322621180

Approach 3: [Contd.]



Training and Validation accuracy and loss plot

	precision	recall	f1-score	support
0	0.92	0.89	0.90	74
1	0.00	0.00	0.00	106
2	0.00	0.00	0.00	98
3	0.74	0.83	0.78	102
4	0.00	0.00	0.00	105
5	0.00	0.00	0.00	38
6	0.02	0.02	0.02	96
7	0.04	0.03	0.03	105
8	0.00	0.00	0.00	40
9	0.00	0.00	0.00	68
10	0.00	0.00	0.00	2
11	0.00	0.00	0.00	100
micro avg	0.17	0.17	0.17	934
macro avg	0.14	0.15	0.15	934
weighted avg	0.16	0.17	0.16	934
samples avg	0.17	0.17	0.17	934

Precision, Recall, and F-1 scores on test set

04

FUTURE SCOPE & CONCLUSION

Final thoughts and explanation of our achievements



Future Scope

- As image processing models improve, better models can be used to execute the task and achieve better results across all classes.
- More images can be added in the dataset to help make the distribution of images in all classes uniform.
- With further improvements in technology, our task of image classification can be combined with all other techniques used to understand lung cancer to build a more robust system.
- The pre-trained model Resnet could be trained for more number of epochs to yield better results on test dataset.

Conclusion

- We were able to identify and classify images into their respective classes that are potential threats when an individual is affected by lung cancer.
- The results obtained are limited to the dataset available on Kaggle to achieve an overall accuracy of 84% using our CNN architecture.
- Pre-trained architectures despite being trained on a very different architecture can translate to a completely different dataset very well if they are trained for a good number of epochs.